Online Appendix to:
Cheap Signals, Costly Proof:
The Reach and Limits of Award-Layer Screening in Cartel Enforcement

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Appendix Roadmap

The appendix builds from definitions to validation. Appendix A fixes the objects: the award-record universe, the CADE–BEC matching rules, the 651-firm adjudication-anchored cobidder label, and the funnel that reconciles every sample count. Appendix B develops the reduced-form ranking model behind the screen—the monotone-likelihood-ratio result that orders firms by loss intensity, the over-crediting logic that flows from it, and the enforcer's optimal-stopping rule that prices how far down the ranking to descend. Appendix C is the core of the validation, where the opportunity-adjusted, timing, leakage, and case-composition audits discipline what the ranking can and cannot mean. Appendix D characterizes the flagged firms and reports the compact score diagnostics; Appendix E adds the bid-layer benchmark and traces the cost–recall frontier of sequential award-to-bid gatekeeping. Appendix F reports the price-scope evidence and the adaptive-use diagnostics, and Appendix G re-runs the validation battery on the federal ComprasNet panel as an out-of-platform stress test. Full grids, permutation draws, threshold sweeps, the bid-feature dictionary, the long proofs, and the supporting figures appear in the Online Supplement.

One convention governs the whole appendix. The ranking is a statement about *forensic priority*, not a finding of conduct or cartel membership; the validation labels are adjudication-anchored *exposure* measures, not legal findings; and the price evidence describes *scope*, not damages or a causal price effect. We restate this boundary only where a result could otherwise be misread.

Appendix A. Data Construction and Validation Labels

The screen and the labels it is validated against come from different sources kept deliberately apart. The screen is built from procurement award records alone; the validation labels come from CADE adjudications and the BEC co-bidding environments around the adjudicated defendants. The exercise that follows asks whether a cheap award-layer statistic concentrates firms near those defendants, while leaving the question of legal membership entirely to the case record.

Appendix A.1. Award-Record Universe

The BEC procurement universe covers 2009–2019 and contains 1,654,447 tender-item records before the final price-regression restriction; the main price sample contains 1,654,401 observations. For the screening exercise the unit is the firm: a firm enters the outer screening universe if it appears as a participant in at least one BEC tender during the sample window, giving 41,444 participating firms.

Appendix A.2. Screen Construction

For each firm i we construct two award-layer variables, T_i (number of BEC tender-items in which firm i participates) and W_i (number of BEC tender-items won by firm i). The always-loser stratum is the set $\{i : W_i = 0\}$ and contains 16,843 firms. The continuous screening score is $s_i = \log(1 + T_i)$ within that stratum, and the discrete frequent-loser flag is

$$FL_i = \mathbf{1}\{W_i = 0,\ T_i \geq 14\}.$$

The threshold comes from the always-loser participation distribution—median plus 1.5 times the interquartile range, with statistical cutoff 13.5 and integer rule $T_i \geq 14$—and is fixed from that distribution alone, not estimated from CADE labels. It flags 2,735 firms, or 16.2% of the always-loser stratum. Because the threshold is pre-specified, the validation exercise asks a genuine question: whether an award-layer rule concentrates firms later observed near adjudicated cartel defendants.

Appendix A.3. Matching CADE Defendants and Constructing Cobidder Labels

CADE supplies 65 firm-defendants across 12 adjudicated procurement-cartel cases with BEC subject matter. We match defendants to BEC records by exact equality on the

Table A.1: Screen Construction

Object	Value	Construction role
BEC participants	41,444	Outer firm universe for award-record construction
Always-losers	16,843	Zero-win stratum in which the screen ranks firms
Participation threshold	14	Integer rule from median $+1.5\times$IQR
Frequent losers	2,735	Discrete first-stage flag within always-losers
Continuous score	$\log(1 + T_i)$	Rank used when a threshold is not required

full 14-digit CNPJ (*códigofornecedor*), zero-padded to preserve leading zeros and with no fuzzy, name-based, or manual step. Of the 65, 47 appear in at least one BEC tender-item during the window and form the direct-defendant set; the rest operate outside BEC or lack matchable identifiers. Matching the defendants to tender-items also locates the adjudication-anchored exposure set, and the $47 \to 48 \to 41$ reconciliation across the crossmatch and participation tables is traced in Online Supplement S7.

The validation target is not the defendant but the *always-loser cobidder*, fixed by one rule: a unique always-loser firm (win rate = 0 over 2009–2019) that shares at least one BEC tender-item with a BEC-active direct defendant. Three exclusions sharpen it—defendants are dropped from the cobidder set, so the label measures loser-side exposure *around* the anchors; the unidentified bidder code `-1` is dropped; and firms appearing across cases or under duplicate CNPJ are counted once. The rule yields 651 cobidders over the full 12-case portfolio.

The frequent-loser flag plays no part in building this label—it is the award-layer score evaluated *against* it. Of the 651 positives, 341 happen to be frequent losers and 310 are not, a description rather than a restriction. The label is adjudication-anchored exposure, not legal membership: it records that a firm bid on the losing side of tenders involving adjudicated defendants, not that it was itself a defendant or filed a cover bid in any given tender.

Appendix A.4. Conservative and Timing Benchmarks

The conservative benchmark restricts CADE to adjudications judged on or before 2020-12-31, trading sample size for freedom from using later case resolution to define the target. Under the *same* cobidder rule this leaves 4 cases, 19 crossmatch direct defendants, and 208 cobidders, of which 107 are frequent losers. The cut uses judgment dates because conduct-onset dates are unavailable (only 9 of 12 cases are dated; the 3 undated cases are ordered last).

The temporal holdout scores participation from 2009–2016 and evaluates discrimination against labels realized in 2017–2019. These are *retrospective adjudication anchors*: judgments post-date the bidding and we claim no real-time legal knowledge at the scoring date. The split follows enforcement logic, not statistical convention—when an agency uses the award layer it does not yet hold the completed case file. Among the 651 positives, 498 are rankable incumbents with pre-2017 history and 153 are entrants with no history to rank (23.5% of the target). The design is conservative in scoring before the target period but imperfect because adjudications often post-date multi-year conduct; it is a retrospective out-of-time discrimination exercise, not causal prediction or field deployment. Per-case judgment dates and benchmark membership appear in Online Supplement S7.

Appendix A.5. Validation Labels and Sample Reconciliation

Table A.2: Validation Labels

Label	Count	Interpretation
Direct CADE defendants in BEC	47	Adjudicated defendants from procurement-cartel cases (legal anchors)
Always-loser cobidders	651	Always-losers (win rate = 0) sharing a tender-item with a direct defendant; main adjudication-anchored exposure target (not membership)
of which frequent losers	341	Composition of the positives (the FL flag does not define the label)
of which non-frequent-losers	310	Composition of the positives
Conservative cases	4	Pre-2020 adjudication benchmark
Conservative direct defendants	19	Crossmatch defendants in the conservative benchmark
Conservative cobidders	208	Always-loser cobidders in the conservative benchmark (same definition)
Conservative frequent-loser cobidders	107	Frequent losers inside the conservative cobidder target

The direct-defendant and cobidder labels stay separate—the first a legal anchor, the second an exposure label tailored to the loser-side ranking. Every benchmark in the paper uses the *same* cobidder rule; counts differ only because the candidate pool, case window, or common-support restriction differs. The 208 conservative count is the same rule applied to the 4 pre-2020 cases, hence a strict subset of the 651 main target ($107 < 341$ frequent-loser cobidders): a narrower case window can only remove cobidders, never add them. Table A.3 traces how the positive count moves as the candidate pool changes while the definition is held fixed—the 651 positives are stable, and the figures that differ (498 rankable incumbents, 153 entrants, 208 conservative) are common-support, timing, or case-window subsets, not rival definitions.

One reading caution and one design point close the appendix. The bid-feature common-support pool of 16,731 firms is an evaluation sample where the Imhof-style bid moments are complete; it retains all 651 positives and is not a label tally, whose exact rows are those of Table 2. And the award-layer screen uses no bid amounts, ranks, within-tender dispersion, reference prices, LANCES timestamps, CADE outcomes, or price residuals—these enter later only as validation or forensic benchmarks, which keeps the first-stage statistic computable in award-only environments and leaves the bid layer as an independent forensic stage.

Table A.3: Positive-count concordance across validation exercises.

Analysis	Candidate pool	Support restriction	Pos.	Pool N	Why the count differs
Main label funnel (Table 2)	Always-losers (16,843)	none	651	16,843	Reference definition; full 12-case portfolio
Opportunity-adjusted validation	Exposed always-losers (common support)	exposure common support (E_i defined)	651	6,040	Exposed subsample retains all positives; denominators shrink to common support
Strict timing, full test universe	All BEC firms scorable at screening date	score frozen 2009–2016; entrants at score zero	651	41,444	Universe widens to every scorable firm; 153 positives are unrankable entrants
Strict timing, rankable incumbents	Firms with pre-2017 history	pre-window participation > 0	498	32,682	Entrant positives (153) excluded: no pre-window history to rank
Strict timing, training always-loser pool	Always-losers in training window	train-window AL status	651	21,819	Pool defined by train-window zero-win status
Bid-feature common support	Always-losers with complete Imhof moments	complete bid-feature support	651	16,731	112 AL firms lack priced-bid moments; all positives retained
Cost-recall frontier	Bid-feature common-support pool	same as bid benchmark	651	16,731	Same pool; cost denominators added
Conservative pre-2020 benchmark	Always-losers (16,843)	case-window restriction only	208	16,843	Fewer anchor cases under the SAME definition (subset of 651)

Notes. The target definition is identical in every row (broad always-loser cobidder; direct defendants excluded; frequent-loser status never used). Positive counts differ only because later exercises impose common-support, timing, or bid-feature restrictions on the candidate pool.

Appendix B. Evidence-Allocation Framework and Legal Scope

Appendix B.1. The pre-proof ranking problem

The model is deliberately spare: it justifies an ordering rule for scarce forensic attention, not the internal organization of a cartel (formation, side payments, leniency, and damages are treated in Asker, 2010, and Marshall and Marx, 2012). The narrower object is the pre-proof enforcement problem—why a firm that repeatedly loses while still appearing in procurement records can be a useful target for bid-level follow-up.

In a cartel-targeted tender one firm is assigned the winning role and submits the designated bid b^*; a losing-role participant submits $b_\ell > b^*$ and is not meant to win. Winning by deviation creates an internal sanction $\kappa > 0$ (lost future rents, relational standing, or capacity-allocation rights); let r be the private gain from deviating into the winner role. Firms outside the winning role are of two reduced-form *model* types—type C, cartel-deployed losing-role participants, and type G, ordinary losing participants—used only to state the ranking condition; C and G are not observed classes. The screen conditions on the observed event $W_i = 0$ (W_i is the firm's number of wins) and ranks firms by participation count T_i.

Lemma 1 (Zero-win losing role). *If the internal sanction from winning in a losing role exceeds the private deviation gain, $\kappa > r$, a losing-role participant avoids winning: it submits $b_\ell > b^*$, so $\Pr(W_i > 0 \mid C) = 0$ over tenders in which the firm is deployed in the losing role.*

Proof. Deviating to a winning bid gains r but pays κ, unprofitable when $\kappa > r$; the inequality fixes the win probability of the losing role, not the numerical bid. □

The inequality $\kappa > r$ supports conditioning on the zero-win event but does *not* imply that all zero-win firms occupy a losing role: the conditioning set is intentionally broad, since many ordinary firms also never win. The evidentiary content comes from the second observable—participation intensity—and from the dynamic margin governing who remains in the zero-win set.

Appendix B.2. Dynamic participation, exit, and the ranking statistic

The ranking statistic is informative only if persistent zero-win participation is more characteristic of a deployed losing role than of an ordinary losing firm, and we obtain that property *endogenously* from an exit margin. An ordinary zero-win firm G pays cost $c_G > 0$, earns gross surplus $m_G(H_{it})$ that weakly declines in its loss history H_{it}, and continues while

$$V_G(H_{it}) \;=\; m_G(H_{it}) - c_G + \beta\,\mathbb{E}[V_G(H_{i,t+1}) \mid H_{it}] \;\geq\; 0, \tag{B.1}$$

exiting otherwise. A cartel-deployed firm C values continuation through a role value $a_C(H_{it})$ that does not erode with the loss count, so after enough zero-win periods $a_C - c_C > m_G - c_G$ and C stays while G is selectively removed. The consequence is a *selection* result: among surviving zero-win firms

$$\lambda_C \;>\; \lambda_G \qquad \text{on the surviving conditioning set } W_i = 0, \tag{B.2}$$

a wedge derived from differential exit, not posited as an exogenous type effect. The value-function derivation is in Online Supplement S6.2.

Assumption 2 (Role-value persistence). *Among surviving zero-win firms, cartel-deployed losing-role participants have weakly greater net continuation value from repeated losing than ordinary zero-win firms after sufficiently many losses: there is a loss count $\bar{L}$ such that for histories with accumulated losses exceeding $\bar{L}$, $a_C(H_{it}) - c_C \geq m_G(H_{it}) - c_G$.*

Conditional on survival and zero wins, let participation counts follow a Poisson approximation,

$$T_i \mid \tau_i, q \sim \text{Poisson}(\lambda_q \tau_i), \qquad q \in \{C, G\}, \tag{B.3}$$

with τ_i the observed exposure length; (B.2) supplies the ordering $\lambda_C > \lambda_G$ on this surviving set.

Proposition 3 (Persistence ranking). *Suppose (1) losing-role firms avoid winning when $\kappa > r$ (Lemma 1); (2) ordinary zero-win firms exit when continuation value falls below the outside option ((B.1)); (3) role value keeps C in the risk set longer, so the selection wedge (B.2) holds (Assumption 2); and (4) among survivors effective participation intensity is higher for C, $\lambda_C > \lambda_G$. Then the posterior $\Pr(C \mid T_i = t, W_i = 0, \tau_i)$ is non-decreasing in t, and T_i and $\log(1 + T_i)$ are monotone ranking statistics for forensic priority within the zero-win stratum.*

Proof. By Bayes' rule the posterior odds of C against G are proportional to the Poisson likelihood ratio $\exp[-(\lambda_C - \lambda_G)\tau_i](\lambda_C/\lambda_G)^t$, which under $\lambda_C > \lambda_G$ is increasing in t (monotone likelihood ratio of the Poisson family; Karlin and Rubin, 1956), so the posterior is non-decreasing in the count and $\log(1 + T_i)$ preserves the ranking. The full posterior-odds derivation is in Online Supplement S6.1. □

Condition (4)—the surviving participation gap—is the key empirical content, validated against adjudication-anchored labels in §§4–6. This is a ranking result under maintained behavioral conditions, not an identification claim. The main-text binary rule (flagging firms above the median plus 1.5 times the interquartile range of zero-win participation) is an operational coarsening of this ranking; the threshold is not a structural parameter, and agencies could instead use the continuous score, a top-k list, or a budget-calibrated cutoff.

Corollary 4 (Legal scope). *Under Lemma 1, Assumption 2, and Proposition 3, the frequent-loser statistic orders expected investigative value within the zero-win stratum. It is not sufficient for liability, damages, or cartel membership.*

Proof. Triage requires an ordering of expected investigative value, which Proposition 3 supplies. Liability and damages require different predicates—proof of agreement, conduct in particular tenders, and a measure of price injury—that the statistic, carrying only participation and win information, omits. □

Appendix B.3. Empirical survival audit

The exit margin predicts who remains: if role value sustains cartel-deployed losing participation while ordinary zero-win firms are selectively removed, the most exposed firms should persist longest. On a firm-year panel of the 16,843 always-losers (101,366 firm-years), the pattern runs as predicted. FL cobidders persist longer than FL non-cobidders and non-FL always-losers (mean active years 3.51 vs. 2.87 vs. 1.61), and a discrete-time hazard model puts the exit-hazard ratios at 0.231 and 0.365 relative to non-FL always-losers—both well below one. The direction holds under alternative exit definitions, dropping late cohorts, dropping the largest CADE case, and a timing-disciplined to-date label (ratio attenuates to 0.768, still below one). Because the labels are *full-sample*, the exercise is retrospective and mechanism-supporting at most, and "survival" means continued BEC participation, not legal-entity survival. The summary table, Kaplan–Meier curves, full hazard and Cox tables, and sensitivity battery are in Online Supplement S6.4.

Appendix B.4. Evidentiary scope

The survival result is mechanism-supporting at most; ordinary zero-win firms may exit for many reasons. More generally, even if the model is correct, a high value of $\log(1 + T_i)$ raises the posterior probability of loser-side exposure only relative to other zero-win firms: it does not identify an agreement, assign cartel membership, prove any specific bid was cover, or establish a damages base. Those claims require the richer record. This boundary is the legal reason for the staged architecture: the award layer ranks firms on a statistic that is cheap, observable, and monotone under a transparent behavioral assumption, while the bid layer and case record do the evidentiary work that liability requires. The paper's identification continues to rest on the opportunity-adjusted and timing-disciplined tests of §§4–6.

Appendix B.5. The evidence-acquisition stopping rule

The ranking of Proposition 3 tells an agency *whom to consider first* but not *how far down the list to descend* before the cost of opening one more record exceeds its expected return. We add that decision layer as a minimal optimal-stopping rule that takes the award ranking as given and prices the recovery footprint of descending it, in the costly-audit tradition (Becker, 1968; Townsend, 1979; Mookherjee and Png, 1989; Polinsky and Shavell, 2000). It is reduced-form on payoffs and makes the cost–recall frontier of §7 the empirical image of a solved problem.

Let the score order the N always-loser candidates, with $\rho(k)$ the *marginal recovery density* at depth k (expected adjudicable cases recovered by opening the k-th firm, the slope of the award-recall curve), $c > 0$ the per-firm forensic cost indexed to bid rows, $V > 0$ the recovery payoff per case (so only c/V matters), and $B > 0$ a hard budget with shadow price μ. Because the ranking is informative, $\rho(k)$ is non-increasing—the only behavioral assumption, inherited from Proposition 3. The agency chooses stopping depth K to maximize net recovery

$$\max_K \; \Pi(K) \;=\; V\int_0^K \rho(k)\,dk \;-\; c\,\Phi(K), \qquad \text{s.t. } \Phi(K) \le B, \tag{B.4}$$

where $\Phi(K) = \int_0^K \phi(k)\,dk$ is cumulative forensic cost in bid rows and $\phi(k) = \Phi'(k) > 0$ the marginal cost.

Proposition 5 (Optimal stopping depth). *Suppose the marginal-recovery density $\rho(k)$ is non-increasing in rank (inherited from Proposition 3) and the marginal forensic cost $\phi(k) > 0$ is non-decreasing. Then $\Pi(K)$ in (B.4) is concave, and the unconstrained optimum K^* is the unique tangency*

$$V\,\rho(K^*) \;=\; c\,\phi(K^*) \qquad \Longleftrightarrow \qquad \frac{\rho(K^*)}{\phi(K^*)} \;=\; \frac{c}{V}\,: \tag{B.5}$$

the agency descends the ranking until the marginal case recovered per unit forensic cost falls to the cost–value ratio c/V. The optimum $K^*(c/V)$ is non-increasing in c/V, and under a binding budget the optimum is the largest K with $\Phi(K) \leq B$. Sweeping c/V over $(0, \infty)$ *and reading off* $(\Phi(K^*), \text{recall}(K^*))$ *traces the cost–recall frontier of §7.*

Proof. The benefit $V \int_0^K \rho$ is concave because ρ is non-increasing and the cost $c\,\Phi$ is convex because ϕ is non-decreasing; hence Π is concave and the first-order condition $V\rho(K) = c\phi(K)$ is necessary and sufficient for the interior optimum, giving (B.5). The optimum is non-increasing in c/V, and the budget-constrained problem reaches the same locus from the constraint side, with μ playing the role of c. The full concavity and comparative-statics derivation is in Online Supplement S6.3. $\square$

The proposition is anchored to the *award*-recall curve, where monotonicity is inherited from Proposition 3; the two-stage award→bid rerank need not be monotone in the survivor pool, a second-order property treated in Online Supplement S6.3. Two consequences organize the empirical reading. The absence of a single optimal cutoff is a *result*: the optimum is budget-dependent, so the object to report is the frontier—the locus of budget-dependent optima—not one point, and we do not calibrate c/V because agency budgets are unobserved. And because Φ is in bid rows while survivors are the highest-participation firms, marginal forensic cost rises steeply where recovery is richest, so the firm- and bid-row-denominator frontiers diverge. The optimum is bounded by the joint full-observability ceiling, and the rule prices recovery of adjudicable exposure only—never evidence that the score identifies active conduct.

Appendix C. Validation Audits

The validation design addresses three mechanical channels: frequent participants have more chances to appear near any defendant, so raw co-bidding rates are not enough; firms operating in the same procurement environments as CADE anchors meet those anchors more often even under ordinary competition; and item-level labels can reuse the same participation history that constructs the score. We state the logic of each audit and the interpretation it licenses, report the headline numbers, and point to Online Supplement S1 for the full grids behind every audit summarized here.

Throughout, "CADE-defendant contact" is an *adjudication-anchored exposure* label: a firm shares a tender-item with a directly adjudicated CADE defendant. The label records proximity, not liability—it does not make the firm a cartel member or a party to any agreement—and the audits below test forensic priority alone. This legal caveat governs the entire appendix.

Appendix C.1. Opportunity-Cell Construction and Expected Contact

A firm can appear near a CADE anchor simply because it operates in environments where defendants are active; the opportunity cells discipline this exposure before any score is credited. We partition the award layer into procurement-opportunity cells g and benchmark each firm against others facing the same cells, using three nested definitions: **COARSE** = item group (two-digit item-code prefix) $\times$ year; **MEDIUM** = item group $\times$ year $\times$ buyer (the reference for the headline within-stratum results); and **STRICT** = item code $\times$ year $\times$ buyer. For firm i, with n_{ig} the tender-items held in cell g and p_g the leave-one-out cell defendant-contact rate, *expected* contact is $E_i = \sum_g n_{ig}\, p_g$, *observed* contact O_i is the firm's count of defendant-sharing tender-items, and *excess* is $X_i = O_i - E_i$ (standardized $Z_i = X_i/\sqrt{\sum_g n_{ig}\, p_g(1-p_g)}$). The mean cell contact rate is 0.017–0.019; most of the level of O_i is explained by E_i, so exposure does the heavy lifting and excess is the residual the score must rank. The cell-construction, support-retention, and excess-by-tier grids are in Online Supplement S1.1.

Appendix C.2. Opportunity-Adjusted Validation

The control-function audit asks whether the screening score still ranks defendant contact once opportunity exposure E_i is in the model. A ladder of logits and rank-tests (S0–S6) on the always-loser firms is reported in Table C.1: raw baselines including an exposure-only model (S0), score-plus-log(1+E) at each cell granularity (S2), breadth and fixed-effect controls (S3–S4), the binary excess target $\mathbf{1}[X_i > 0]$ (S5), and the strict-support subsample (S6).

The honest reading is deflationary. Exposure alone already classifies cobidder contact well (AUC 0.905 unconditionally; 0.713 on the exposed subsample), and the combined AUCs near 0.98 overstate the score's marginal value because E_i encodes the label. Once exposure is held fixed the residual ordering is small and not robust: the nested DeLong increment is +0.010 ($p = 0.013$)—statistically real but a one-percentage-point gain that would not reorder a forensic queue, and the only test in the family that survives a

Table C.1: Control-Function Validation (Selected Specifications)

Spec	Model	N	Pos.	ROC AUC	PR-AUC
S0	raw score ($\log T$)	16,843	651	0.761	0.143
S0	raw FL14	16,843	651	0.688	0.097
S0	exposure-only (all AL)	16,843	651	0.905	0.300
S2	score $+\log(1{+}E)$ [MEDIUM]	16,843	651	0.985	0.708
S2	FL14 $+\log(1{+}E)$ [MEDIUM]	16,843	651	0.985	0.706
S2	exposure-only logit (exposed)	6,040	651	0.713	0.300
S3	score $+E+$ breadth	16,843	651	0.983	0.711
S4	score + opp-FE + part-FE (exposed)	6,040	651	0.707	0.255
S5	score ranks $\mathbf{1}[X_i{>}0]$ (MEDIUM)	16,843	475	0.764	0.095
S5	FL14 ranks $\mathbf{1}[X_i{>}0]$ (MEDIUM)	16,843	475	0.687	0.073
S6	score $+E$, strict support	7,683	446	0.969	0.671
NESTED	exposure-only logit	6,040	651	0.713	—
NESTED	exposure + score	6,040	651	0.723	—
WITHIN	score | opportunity stratum	6,040	651	0.471	—
WITHIN	FL14 | opportunity stratum	6,040	651	0.507	—

Notes: Outcome is adjudication-anchored cobidder contact (S5 uses the binary excess target $\mathbf{1}[X_i > 0]$). **The combined exposure-plus-score logits in S2–S3 reach AUC ≈ 0.98, but this is mechanically inflated and is NOT evidence of score signal:** the exposure term E_i is built from the same defendant-contact rates that define the label, so a model containing E_i partly predicts the label by construction. Holding exposure fixed, the residual ordering is statistically detectable but operationally negligible: the nested DeLong increment over the exposure-only model is only $+0.010$ AUC ($p = 0.013$), while the within-opportunity-stratum AUC for the score is 0.471 ($\approx$ chance; FL14 0.507). The companion permutation and matched audits in Online Supplement S1.2 do not establish a robust residual.

Benjamini–Hochberg correction— while the within-stratum AUC is at chance (0.471 for the score, 0.507 for FL14). The three companion exercises in Online Supplement S1.2—a matched-stratum permutation ($p = 0.127$), a firm-exposure simulation ($p = 1.000$), and a CEM opportunity-matched comparison (within-stratum AUC 0.626)—do not corroborate a robust residual, and the within-stratum change in Pr(cobidder) for FL14 versus non-FL is in fact *negative* (-0.017). We therefore report no operationally relevant residual ordering net of opportunity.

Audit of the audit.. Because a negative verdict can be manufactured as easily as an inflated one, we turn the audit on its own machinery; the full leakage-tier, falsifiability, power, and label-frozen-timing detail is in Online Supplement S1.3. The headline is one number: a label-blind opportunity ranking—recomputing every cell's defendant density from non-cobidder rows only, so no eventual positive contributes to any firm's expected contact—gives AUC 0.553, confirming that the exposure benchmark is an upper bound on opportunity's role and the within-stratum floor is the genuine signal. Three supporting checks in the supplement (a falsifiable positive control, a calibrated power analysis that rejects with probability 0.97 at a true within-stratum AUC of 0.55, and a label-frozen 2009–2016 design) all reinforce that no cartel-specific residual survives the volume channel.

Appendix C.3. The Over-Crediting Bias as an Estimable Object

The opportunity audit shows *that* the raw award score over-credits defendant contact; this subsection characterizes *why* and turns the caveat into a diagnostic a practitioner can compute before opening a single bid file. The cobidder label is contact-defined—a firm is a positive if it shares at least one tender-item with an adjudicated defendant—and the probability of such a contact is mechanically increasing in participation volume T_i,

$$\Pr(\text{cobidder}_i = 1 \mid T_i) \;=\; 1 - (1-\pi)^{T_i} \;\approx\; \pi\, T_i \qquad (\pi T_i \text{ small}), \tag{C.1}$$

where π is the per-tender-item defendant-contact rate (the cell density p_g of Appendix C.1). The award score is monotone in the same T_i ($\log(1{+}T_i)$), so *both the score and the label load on participation volume*: the score orders the label not because loss intensity is behaviorally informative but because each is a function of T—the analytic shape of the within-stratum null (0.471, $\approx$ chance) and the label-blind exposure ranking (0.553).

Write F_T for the distribution of T in the candidate pool. Under the rare-target linearization the positive class is the *size-biased* T (weighted by πT) and the negative class is approximately F_T; since the score is monotone in T, the raw ROC area is the size-bias stochastic gap

$$\text{AUC}_{\text{raw}} \;=\; \Pr(T_i > T_j), \qquad T_i \sim \widetilde{F}_T \text{ (size-biased)}, \;\; T_j \sim F_T. \tag{C.2}$$

The opportunity-adjusted area $\text{AUC}_{\text{opp-adj}}$ holds the volume channel fixed; it is the genuine-signal floor, which the audits put at $\approx$ chance. The over-crediting bias is the gap $\Delta = \text{AUC}_{\text{raw}} - \text{AUC}_{\text{opp-adj}}$.

Proof of Proposition 1 (over-crediting signs). (a) For a positive i size-biased by weight $w(T) = \pi T$ and a negative $j \sim F_T$, $\Pr(T_i > T_j) - \frac{1}{2}$ equals the rank covariance between $w(T)$ and the indicator that T exceeds an independent copy; this covariance is non-negative and increasing in the spread of F_T, vanishing when F_T is degenerate. For a scale family the spread is summarized by $\text{CV}(T)$, and the size-bias gap $\mathbb{E}[T^2]/(\mathbb{E}[T])^2 - 1 = \text{CV}^2(T)$

is monotone in it. (b) Differentiating $1-(1-\pi)^T$ in T gives a marginal contact response $-(1-\pi)^T \log(1-\pi)$ that flattens in T as π grows; the weight $w(T)$ thus becomes less steeply increasing, the positive class less size-biased, and the gap in (C.2) smaller. Both statements concern the *signs* only; the magnitude is not closed-form once one leaves the linearization and uses the empirical heavy-tailed F_T, and is read from the simulation below. □

We state *signs*, not a magnitude: off the linearization a closed-form formula would be spurious precision, so we characterize the gap by a transparent simulation. Figure C.1 displays a *synthetic surface anchored at one empirical point per platform*, not an estimated inflation curve: it draws T from a Gamma family indexed by $\mathrm{CV}(T)$, assigns the contact label by (C.1) at a target base rate, scores firms by a noisy $\log(1+T)$ calibrated once so the BEC coordinate reproduces its observed raw area, and compares the raw ROC area to the within-T-stratum area. No confidential microdata enter: the only empirical inputs are two published scalars per platform—the coefficient of variation of always-loser participation counts and the adjudicated base rate—which fix the calibration anchor and the two overlaid platform points; everything else is synthetic backbone. Across the grid the surface confirms both signs of Proposition 1: Δ rises monotonically in $\mathrm{CV}(T)$ (from ≈ 0.03 at $\mathrm{CV} = 0.5$ to ≈ 0.34 at $\mathrm{CV} = 5$) and falls in the base rate once the target leaves the rare regime.

The leading-order sufficient statistic.. The practitioner deliverable is that $\mathrm{CV}(T)$ of the candidate pool—computable from award data alone before any bid file is opened—together with the target base rate bounds how much of any raw area is mechanical. High dispersion of participation volume warns that a contact-anchored validation will over-credit a volume-loaded score; the correct response is the within-stratum / label-blind comparison, not the raw area. This is a *diagnostic*, not a fix: the correction reveals the genuine floor to be $\approx$ chance (Appendix C.2); it does not rescue the screen. The two platforms are a second coordinate on the same surface: BEC ($\mathrm{CV}(T) = 2.79$, base rate 3.9%, $\Delta \approx 0.29$) and the federal platform ($\mathrm{CV}(T) = 3.79$, base rate 0.5%, $\Delta \approx 0.28$). The rarer federal target is why its exposure-only model *edges past* the raw score ($0.754 > 0.744$): at a rarer target the size-biasing is nearly complete. The inflation *law* replicates at full power across both platforms; only the residual within-stratum floor is power-bounded federally (below an

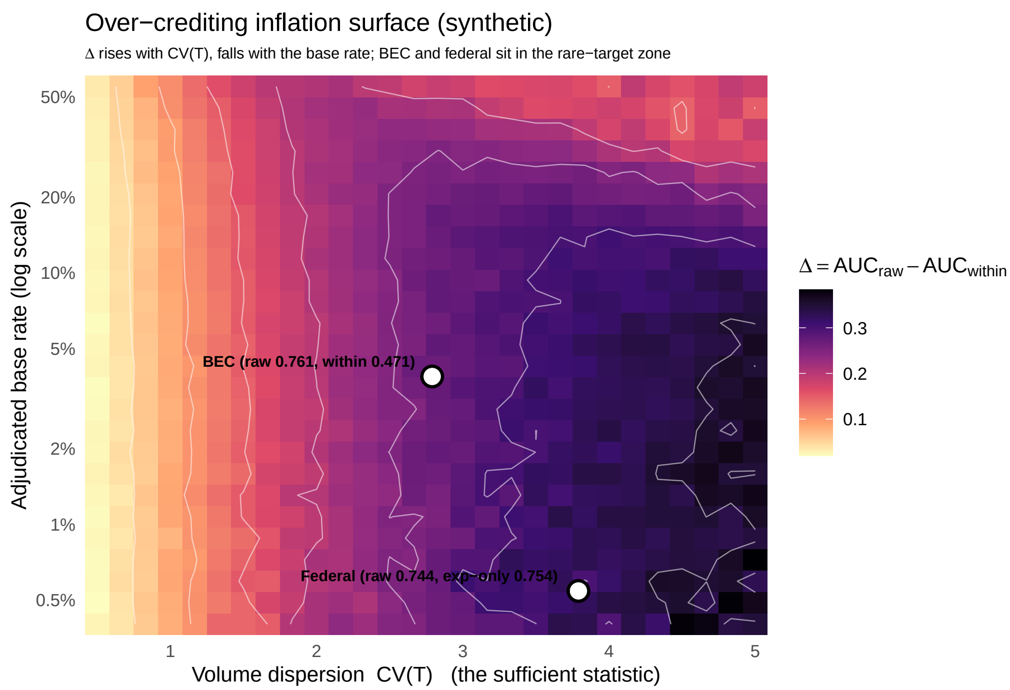

Figure C.1: Over-crediting bias $\Delta = \text{AUC}_{\text{raw}} - \text{AUC}_{\text{opp-adj}}$ over the volume-dispersion $\times$ base-rate grid. This is a *synthetic surface anchored at one empirical point per platform*, not an estimated inflation curve: the backbone is simulated from a Gamma family, calibrated once so the BEC coordinate reproduces its observed raw area, and the two real platform points are then *overlaid* at their measured $(\text{CV}(T), \text{base rate})$ coordinates—BEC ($\text{CV} = 2.79$, base rate 3.9%) and federal ComprasNet ($\text{CV} = 3.79$, base rate 0.5%). The surface increases in $\text{CV}(T)$ and decreases in the adjudicated base rate, as Proposition 1 predicts; federal sits in a higher-Δ region because higher dispersion and a roughly seven-fold lower base rate both push toward more complete inflation. Only $\text{CV}(T)$ and the base rate are taken from data.

AUC of roughly 0.55 at $N^+ = 195$).

Appendix C.4. Timing Information Sets and Strict Holdout

This block holds *time* fixed: could a regulator who froze the score *before* the evaluation window have ranked the firms later adjudicated as cobidders? The CADE files carry only judgment dates (mostly 2015–2025) and *no conduct dates*, so all timing is *year-level* (BEC participation year), never day-level (`DAY_LEVEL_TIMING_UNAVAILABLE`). We define seven candidate information sets, A–G, and lead on the honest incumbent-triage designs C/D/G (strict prospective score, rankable incumbent pool, retrospective label); the full A–G register and the leakage-channel cross-tabulation are in Online Supplement S1.4. The strict holdout freezes the score, always-loser status, and median+1.5×IQR threshold on 2009–2016 (train-window threshold 7, vs. full-window 13.5) and evaluates against 2017–2019 cobidder labels. Table C.2 gives five nested samples.

Table C.2: Strict Holdout 2009–2016 → 2017–2019 (Year-Level Timing)

Sample	N	Pos.	ROC AUC	PR-AUC	Prec@500	Recall@500	Lift@500	Tie-at-0
1. Full test candidate	41,444	651	0.474	0.014	0.000	0.000	0.00	0.246
2. Rankable (T^{train}>0)	32,682	498	0.471	0.013	0.000	0.000	0.00	0.043
3. Training always-loser	21,819	651	0.684	0.077	0.124	0.095	4.16	0.402
4. Training AL & test-active	11,351	312	0.667	0.084	0.142	0.228	5.17	0.772
5. Zero-win both (LEAKS)	16,843	651	0.646	0.086	0.136	0.104	3.52	0.280

Notes: Continuous loser-side score ($\log T^{\text{train}}$). Sample 3 is the honest strict pool; within it, the FL-binary AUC is 0.646 and the continuous AUC is 0.684 (threshold = 7, vs. full-window 13.5). Sample 5 leaks future wins via full-period always-loser status and is reported as a leak-flagged sensitivity only. **On the full candidate universe (sample 1) the strict screen collapses to chance:** ROC 0.474 (below chance), PR-AUC 0.014, and precision@500 =recall@500 = 0, with about 25% of firms tied at score zero. Of the 498 rankable positives and 153 entrant positives (23.5% of 651), entrants form a structural blind spot the prospective score cannot reach. "Tie-at-0" is the share of the candidate set carrying score zero. Source: `table_D_strict_2009_2016_to_2017_2019.csv`; `strict_holdout_composition.csv`.

The reading is deliberately cautious. Only a limited within-pool residual ordering remains (sample 3, AUC 0.684–0.646), but on the full firm universe the strict screen has no operational traction (precision@500 = 0): the entrant coverage hole and the mass of score-zero ties are the proximate causes. Two companion exercises in Online Supplement S1.4 confirm the pattern—a rolling-origin validation posting precision@500 = 0 on the full universe in every origin year, and a leakage decomposition leaving the surviving structural component in the 0.86–0.89 range under cobidder-firm holdout (out-of-fold AUC 0.891) and temporal holdout (AUC 0.864).

Appendix C.5. Case Composition, Scope, and Limitations

The pooled signal is concentrated in a few adjudicated cases rather than the construct: dropping the single largest case (`trens_metros`) lowers PR-AUC from 0.143 to 0.090 (-37%), and case-balanced precision@500 is 0.043 against the firm-weighted pooled 0.216, with the largest case holding 32.0% of positives. The leave-largest-case-out grid, concentration measures, and a clustered randomization-inference test (item-group $\times$ year clusters, $B = 1{,}000$: ranking metrics $p = 0.001$; top-500 coverage *not* significant, $p = 0.103$) are in Online Supplement S1.5. Inference rests on a small number of adjudicated cartels (twelve matched CADE cases, six informative), so the clustered p-values are directionally informative rather than exactly size-controlled.

The asymmetry against direct defendants is the design, not an escape hatch. A generic cartel-membership detector would classify direct CADE defendants as well as cobidders; this one does not. The FL-binary score ranks direct defendants at ROC ≈ 0.49 under both full and strict timing (0.491, 0.492)—chance—with precision@500 $= 0$; continuous volume carries some information (ROC 0.66–0.70) but with precision@500 ≤ 0.004, so it does not operationally retrieve them either (full grid in Online Supplement S1.6). The loser-side *binary* flag is uninformative about defendants, who typically sit on the winner side of the arrangement.

Several limitations bound this evidence. The opportunity cells are *award-layer proxies* built from item, year, and procuring-unit identifiers, with modality dropped because it is absent from that layer. All timing is *year-level*: the CADE files carry no conduct dates, so the adjudication-observable design is infeasible rather than estimated. There is an *entrant coverage hole* (153/651, 23.5% of positives carry score zero) and the signal is *concentrated* by case, item group, and year. ROC is a red herring at sub-1% prevalence; precision and recall are the operational metrics, and they stay low under the pooled diagnostic and collapse to zero on the full strict-timing universe. Net of opportunity and timing the residual ordering is *marginal at best and not robust across designs*; what the award layer reliably delivers is a coarse, retrospective basis for prioritizing scrutiny, and it is **not legal proof, not identification of cartel members, and not an estimate of damages or causal effect**.

Appendix D. Economic Profile and Robustness: Summary

The profiles characterize *losing behavior* on the award layer and feed a triage ranking; "cobidder" is an adjudication-anchored exposure label throughout—a firm that co-bid with a firm named in a CADE proceeding—and carries no inference of participation in any agreement. All groups are drawn from the always-loser universe ($N = 16{,}843$ firms with zero recorded wins over the calibration window), which is distinct from the frequent-loser cut: the validation target is the broad set of 651 always-loser cobidders, of which only 341 (52.4%) satisfy the frequent-loser threshold, so the label is not frequent-loser-conditioned. The 47 direct CADE defendants are *winners* and are excluded from every cobidder group. The Section 6 profile contrast runs *within* the frequent-loser stratum—group D (FL non-cobidders, $N = 2{,}394$) against group E (FL cobidders, $N = 341$)—so the estimated gaps cannot be an artifact of the cut. Profiles are computed from award records in five panels (participation/survival, breadth/concentration, defendant-proximity/opportunity, modality-quorum proxies, timing); three classes that would sharpen interpretation—tender value, distance to winner, and fine geography—are unobserved on the award layer, so the ordinary-loser alternatives of Section 6 are narrowed but not eliminated. The full group register, the complete standardized-difference table, the threshold sweep, the bunching histogram, the market-specific zero-win variants, and the negative-control draws are in Online Supplement S2 (groups in S2.1, full table in S2.2, robustness families in S2.3).

Table D.1 reports the top standardized mean differences (Cohen's d, E minus D); we use d rather than p-values, which reject essentially any difference at N in the thousands. The largest gaps are defendant-proximity variables, and they are *partly mechanical*: the cobidder label is itself defined by adjudication-anchored contact, so firms with more contact are more likely to be labeled. Holding exposure fixed—regressing each variable on the cobidder indicator with controls for $\log E_i^{\mathrm{MED}}$, a binned score term, breadth, and active years, and coarsened-exact matching on E_i^{MED} deciles—collapses the item-group HHI gap from $d \approx 0.13$ to ≈ 0 and the breadth and persistence gaps toward zero, so they reflect procurement-environment composition rather than intrinsic firm traits. Only the defendant-contact variables survive, and that survival is partly a construction artifact; it should not be read as independent corroboration of conduct. The score ranks firms by exposure to defendant-adjacent opportunities, not by conduct intensity, and the robustness families in Online Supplement S2.3 corroborate that reading rather than rescue the screen: prevalence rises with participation while excess contact does not, the administrative FL14 cut shows no strategic bunching, and against matched placebo anchors the real-anchor discrimination is indistinguishable from chance ($p = 0.46$). Award-layer profiles describe patterns of *losing*; they are not proof of any agreement.

Table D.1: Standardized profile differences (top variables), E (FL cobidders) minus D (FL non-cobidders).

Panel	Variable	Label	Raw diff.	Cohen's d
C	*n_cases*	# CADE cases touched	1.109	6.169
C	*n_def_firms*	# distinct defendants	1.370	4.270
C	*contact_intensity*	O_i/T_i	0.172	2.119
C	*share_in_def_cells*	part. in def. cells	0.205	1.829
C	O_i	observed contacts	7.658	1.806
C	X_i^{MED}	excess contacts	1.523	1.212
B	*n_buyers*	# distinct buyers	7.831	0.532
A	T_i	# tender-items	24.689	0.438
B	*pregao_share*	Pregão share	0.173	0.436
A	*n_years*	# active years	0.625	0.391
B	*convite_share*	Convite share	−0.173	−0.436

Notes: $n_D = 2{,}394$, $n_E = 341$ (modality shares use $n_D = 2{,}387$ where modality is defined). Positive d: cobidders are higher. Top eleven variables by $|d|$; the full table is in Online Supplement S2.2. The largest gaps are defendant-proximity variables (*n_cases* $d = 6.17$, *n_def_firms* $d = 4.27$), which are partly mechanical because the cobidder label is itself adjudication-anchored. The non-mechanical breadth and persistence gaps are moderate (*n_buyers* $d = 0.53$, T_i $d = 0.44$, *n_years* $d = 0.39$). Item-group HHI shows only a small raw gap ($d = 0.13$) that falls to ≈ 0 after opportunity adjustment.

Appendix E. Bid-Layer Benchmark and Cost-Recall Details

Section 6.1 asks whether the cheap, award-layer participation signal carries ranking information comparable to a transparent bid-moment benchmark built from full bid microdata; this appendix gives the model summary, and the full construction, feature dictionary, fold audits, and design-by-design performance table are in Online Supplement S3. The benchmark is an *implemented full-observability* diagnostic: it computes seven Imhof–Wallimann within-tender bid-distribution moments, aggregates them to the firm, and feeds them to a random forest evaluated by cross-validation. It is neither a state-of-the-art horse race nor a first-stage triage tool, because the moments require every within-tender bid to be observed and recovered. "Cobidder" is an adjudication-anchored exposure label throughout—an always-loser firm that shares at least one BEC tender-item with a BEC-active direct CADE defendant—so the benchmark *ranks* that exposure target for forensic priority and makes no claim to identify cartelists.

Appendix E.1. Data, Units, and Support

The bid layer is built from one row per priced bid (firm, OC sequence, item, `bid_price`), retaining a bid only if `bid_price > 0`. The modeling unit is the *firm*: per-tender moments are shared across all firms bidding on a tender-item and then firm-mean aggregated. Of the 16,843 always-losers, 64 are dropped for having no priced bids or fewer than two bids per tender (so the coefficient of variation, skewness, and kurtosis are undefined). Excluding the direct CADE defendants by upstream label curation leaves an evaluation pool of 16,731 firms, of which 651 are always-loser cobidder positives. **All 651 positive (cobidder) labels are non-missing on every Imhof moment**: the support loss falls entirely on the negatives, so there is no positive attrition. The seven moments cover four families—dispersion, level/relative spread, shape, and rank/order—and require the full within-tender bid record, which the award layer does not contain. The full feature dictionary, units-of-observation and per-feature missingness tables are in Online Supplement S3.

Appendix E.2. Model Specification

The learner is a random forest (`ranger`, 500 trees, `probability = TRUE`), with no hyperparameter tuning and no standardization (the forest is scale-invariant); missingness is handled listwise through the pool restriction. The outcome is binary `is_cade` (1 for an always-loser cobidder in the candidate pool, not a direct defendant). Evaluation is by 5-fold cross-validation, each firm receiving one held-out predicted probability; AUC and DeLong tests are computed on these out-of-fold firm-level predictions. Four designs of increasing stringency are run—pooled-random (a diagnostic upper bound), case-grouped (the honest robustness test), exclude-label-defining-tenders (the contamination check), and environment holdout. Strict bid-timing validation is *blocked*: the within-tender moments cannot be cleanly time-limited without re-deriving them on a per-period bid panel. The full spec-by-spec audit and fold compositions are in Online Supplement S3.

Appendix E.3. Leakage, Contamination, and Performance

The award score is *clean* (a CADE-label-independent rank on `log1p(tenders_count)`). The bid and combined forests under full-period features and random folds are **diagnostic-only**: random folds are optimistic, and the firm-mean moments include label-defining tenders. The contamination is small: excluding label-defining tenders leaves the bid ROC essentially unchanged (0.717 → 0.713), so the fragility is case clustering, not feature–target overlap; the bid ROC itself falls from 0.717 (random) to 0.626 (case-grouped), while the combined model's case-grouped PR-AUC (0.103) falls below award-only (0.143). The headline ranking comparison—the cheap award layer's comparable discrimination at lower cost, case-fragile complementarity, and the moderate award/bid rank correlation—is reported and interpreted in the main text (Section 6.1, Table 5). The full design-by-design performance table, leakage audit, complementarity, and calibration tables are in Online Supplement S3 (bid-layer performance table).

Appendix E.4. Limitations

The bid-layer benchmark is a full-observability diagnostic, not a first stage: the Imhof moments require all within-tender bids, which is exactly the information cost the award-layer screen avoids; the headline pooled-random numbers carry random-CV optimism; strict bid-timing validation is *blocked*, because within-tender dispersion moments cannot be cleanly restricted to a pre-target window; the benchmark is deliberately modest (untuned, unstandardized Imhof–Wallimann moments in a random forest), not a state-of-the-art horse race; and the target is the 651 always-loser cobidders, an adjudication-anchored exposure measure, so false positives are unlabeled firms and nothing here constitutes proof of conduct.

Appendix E.5. Sequential Gatekeeping and the Cost-Recall Frontier

The sequential architecture separates two tasks that are often conflated. The award-layer stage ranks firms using fields available before any bid microdata are recovered; the bid-layer stage then builds richer forensic features only for the firms that survive the first stage, so the comparison reads as an evidence-allocation design rather than a claim that thin records substitute for full bid files. The output is a forensic-priority queue, not a liability finding. Throughout, "cost" denotes a *data-processing footprint*—the firms, tender-items, bid rows, and buyer-item-year cells opened to compute the bid-layer features—never a monetary agency cost, which we do not observe. Discriminating performance is adjudication-anchored: positives are the 651 always-loser cobidder firms that link to a CADE-adjudicated case, against the complete-Imhof pool of 16,772 firms. The operating points are retrospective cost-footprint designs conditional on the validated incumbent ranking, not fully prospective deployment tests. Three scores are defined over the pool—an **award score** $s_i = \log(1 + T_i)$ from routine administrative fields, a **bid score** (a random forest on within-tender bid-distribution moments requiring each opened tender-item's full bid record), and a **combined score**—and seven ranking rules are compared, distinguished by which layer they consume and whether they triage before opening bid microdata. The full rule definitions, the bid-feature dictionary, the random-forest specification, and the four validation designs are in Online Supplement S3.

Appendix E.5.1. Cost Denominators

Bid-layer recovery is costlier at scale because within-tender moments require the *full* set of bids in each opened tender-item, not only the focal firm's: opening a survivor firm means opening every tender-item that firm participated in. We therefore measure the recovery footprint along several denominators rather than a single headline number; Table E.1 defines them.

Table E.1: Cost denominators for the recovery footprint.

ID	Denominator (full-observability value)	Value
D1	Firms opened ($= \lvert\mathcal{S}\rvert$)	16,772
D2	Survivor firm-tender-item rows (with multiplicity)	163,264
D3	Opened tender-items (de-duplicated; recovery unit)	116,112
D4	Opened bid rows, full-tender (processing footprint)	3,393,915
D5	Survivor bid rows only (lower bound)	391,610
D6	Buyer × item-group × year cells touched	58,936
D7	Buyers (PBU codes) touched	1,244
D8	Item-groups (2-digit) touched	90

Source: `full_observability_costs.csv`; denominator definitions in `table_G_cost_denominator_definitions.csv`. Values are the full-pool footprint (K_1 = 16,772); the grid in Online Supplement S3 reports them at each K_1.

D4 is the realistic processing footprint: within-tender Imhof moments require the entire tender record, so D4 counts all bids in every opened tender-item, not only the survivors' own bids (D5).

Not observed (no value computed): LANCES export units (D9); legal or subpoena-style access requests (D10); monetary cost in reais. *Analyst-hours* are reported only as an illustrative processing-burden proxy (D11), never as a measured agency cost.

The firm count (D1) and the bid-row footprint (D4) diverge sharply, and the divergence is the central caution. Survivors are high-participation firms: at the $K_1 = 2000$ operating point an 88% firm reduction opens about 67% of the bid rows—only a 33% bid-row reduction. The firm-level reduction therefore overstates the saving; the tender-item and bid-row denominators are the honest processing-footprint measures.

Appendix E.5.2. Cost-Recall Frontier and Findings

The sequential award $\rightarrow$ bid rule traces a cost-recall frontier that tracks close to the joint full-observability upper bound while opening fewer tender-items but never crosses it ($124/133 = 93\%$ of joint top-500 true positives at $K_1 = 1{,}000$). It sits well above a same-size random survivor floor, so it is the award ranking, not the mere act of opening K_1 firms, that carries the signal: the award-ranked survivor pool captures most positives *before* any bid rerank, so the costly bid features *reorder* survivors into a forensic-priority queue rather than discovering positives the award gate missed. The bid layer's marginal contribution is ordering—an evidence-allocation gain, not a detection gain—and no single (K_1, k) pair dominates, so the frontier is a menu, not a cutoff. The full grid, the frontier figure, the gatekeeping algorithm, and the operating-point logic are in Online Supplement S3. Two qualifications carry the interpretation. A timing-disciplined award gate stays informative (strict-timing AUC 0.684 on the training-AL pool against 0.761 full-window), while the sequential rule's within-tender moments cannot be re-derived on a pre-period basis, so its timing test is reported as infeasible. And performance is case-fragile: the largest case supplies 32.0% of positives and 45.4% of the top-500 true positives. Cost is a processing footprint, not money; labels are limited and false positives unlabeled, so precision is a lower bound on the queue's triage value. The estimated ranking is not a portable cartel score; the transferable object is the decomposition framework.

Appendix F. Price Scope and Adaptive Use

The main text (Section 8) reads the price evidence as corroboration of economic non-neutrality, not as causal identification. We give the price sample, the broad-to-within-cell sign reversal, the high-value exception, the direct-CADE null, and the diagnostics that keep the interpretation on the scope side of the line; the full broad-by-modality and quartile/trim grids live in Online Supplement F. The companion exercise on how a published administrative cutoff degrades under strategic response—an institutional-design probe, not a defense of any behavioral signal—is in Appendix F.6.

Appendix F.1. Price sample and outcome

The outcome is the log negotiated unit price at the item–award level. The estimation sample is the broad item-level panel with $N = 1{,}654{,}401$ awards, with item, year, and (where indicated) procuring-unit (PBU) fixed effects. The overlap, quartile, and trim variants reweight or restrict this sample but do not change the outcome or the unit of observation.

Appendix F.2. Broad associations and overlap-cell sign reversal

In the broad specification, frequent-loser presence is positively associated with the log negotiated unit price (broad coefficient +0.064, headline range $+3.6\% - +7.7\%$ across estimators). These are descriptive associations, not causal effects.

The positive broad coefficient is a weighting artifact, not a robust treatment sign. Holding the overlap cells fixed and moving from broad-sample weighting to the within-cell average treatment effect on the treated (ATT) reverses the sign monotonically: broad +0.064 ($p = 0.003$); overlap-cell unweighted +0.044 ($p = 0.035$); overlap-cell ATT −0.097 ($p < 0.001$); PS-trimmed −0.307. Only 1.06% of treated items lack a within-cell counterfactual, so the reversal cannot be a dropped-observation artifact. Table F.1 reports the progression.

Table F.1: Sign-Reversal Decomposition (overlap-cell reweighting)

Specification	Coefficient	p-value
Broad sample	+0.064	0.003
Overlap-cell, unweighted	+0.044	0.035
Overlap-cell ATT	−0.097	< 0.001
Overlap-cell ATT, PS-trimmed	−0.307	—

Notes: Same outcome and overlap cells throughout; rows differ only in weighting and trimming. The broad-to-ATT move reverses the sign because it strips the between-cell selection component. Only 1.06% of treated items lack a within-cell counterfactual.

Appendix F.3. High-value segment, direct-CADE null, and the scope boundary

The negative ATT reading survives both modalities (Convite −0.099, Pregão −0.098) and every tender-value quartile except the highest; the high-value tail (Q4) is the only systematically positive cell (+0.041, $p = 0.045$), where coordination costs are spread over larger contracts. Trimming the most heavily weighted overlap cells does not eliminate the negative aggregate. The full quartile and trim grid is in Online Supplement S4. Restricting the overlap to items involving firms named as direct CADE defendants yields a null price estimate (-0.061, $p = 0.45$; non-direct overlap -0.097): the price regressions do not recover the legal object of the adjudicated cases. Three facts keep the evidence on the scope side of the line. The broad positive imprint is selection into structurally higher-priced cells (among non-treated items, mean log price rises from 1.35 to 6.93 across cell FL-share quintiles, a $\Delta = 5.58$ log-point gap, partialled-out coefficient +3.55 once all five cell dimensions are absorbed), and the within-cell ATT removes exactly this component. The direct-CADE null means there is no price base tied to the adjudicated conduct. And the design holds the procurement environment fixed rather than against a counterfactual price, so no overcharge can be computed. *These are scope facts about where flagged environments sit in the price distribution, not damages or overcharge.*

Appendix F.4. Within-cell mechanism

The within-cell price compression loads on the count of genuine bidders, not on the frequent-loser count: the within-cell coefficient is -0.048 (SE 0.004) and falls to +0.008 (not significant) once log(n_firms) is added, so the mechanism is *not* identified by these data.

Appendix F.5. Limitations

The price evidence is descriptive, environment-specific, and not causal. It establishes economic non-neutrality of flagged environments, concentrated in the high-value tail, but it is not a damages estimate, an overcharge, a markup, or a treatment effect on price. The coefficients are BEC-specific; another agency should re-validate the ranking against its own enforcement anchors and price benchmarks rather than reusing these estimates or the BEC cutoff.

Appendix F.6. Instability of a Published Cutoff Under Strategic Response (Retrospective)

How quickly does a static, published threshold decay as a triage device once participants can react to it? Because Section 7 finds no conduct-specific residual, the simulations here defend nothing behavioral; they are an institutional-design exercise. Holding the empirical environment fixed, we re-evaluate the participation score under simple evasive responses. The exercise is retrospective, not a fully prospective deployment test: the labels remain adjudication-anchored and the operating points are conditional on the validated incumbent ranking. The verdict is a hierarchy. Rotation of losing-role firms and occasional small wins leave discrimination essentially unchanged (absorbed by the persistence signal); CNPJ splitting erodes it by breaking participation history; threshold-aware capping is the most damaging, and most damaging of all combined with rotation. The mapping to the cost–recall frontier is that a published threshold invites bunching and is the most gameable rule, so continuous ranks or capacity-constrained top-k queues should be preferred over a static published cutoff. The full AUC-by-evasion-mode grid and the simulation detail are in Online Supplement S4. This is a diagnostic, not a full equilibrium model of evasion: it measures how fast a static published cutoff loses its triage value once firms respond to it, and makes no claim about the durability of any underlying behavioral pattern.

Appendix G. Federal Audit Battery (ComprasNet)

The federal ComprasNet panel (2013–2019) is a *stress test* of how far the audit ports across platforms, re-running the BEC validation battery of Appendix C stage for stage under the same definitions; it supplies the federal column of Table 5 in Section 5. As on BEC, "CADE-defendant contact" is an adjudication-anchored *exposure* label—a firm shares a tender-item with a directly adjudicated CADE defendant—and what the battery measures is how far forensic priority *ports*, not membership or liability. The legal anchors are the same numbered CADE cases used on BEC, *partially overlapping* with the BEC anchors by construction, so this is not an independent universe. The federal panel has no sealed-bid modality, so no modality stratification is reported. The full construction—label funnel, opportunity cells, and the complete control-function, timing, leave-one-case-out, negative-control, and armor batteries—is in Online Supplement S5.

Appendix G.1. What the Battery Returns

The federal reading reproduces the BEC pattern on both points of the opportunity-inflation curve. *First*, the raw award-layer concentration looks informative (ROC-AUC

0.744), but once opportunity exposure is held fixed it is opportunity arithmetic: the label-blind opportunity ranking retains 0.611 on its own, the combined exposure-plus-score logit is mechanically inflated because E_i is built from the same defendant-contact rates that define the label, and the score's isolated contribution—the nested DeLong increment over the exposure-only model, +0.005 ($p = 0.191$), and the within-opportunity-stratum AUC 0.462—is small. *Second*, the within-stratum null is power-bounded, not merely underpowered: the within-stratum positive control reaches 0.992 (so the design recovers a residual when one exists), and the matched-strata permutation rejects with probability 0.35 once the true within-stratum AUC reaches 0.55 (probability 0.90 at 0.60), yet the observed PR-AUC does not reject the matched-permutation null ($p = 0.906$). The prospective ranking collapses as it does on BEC, and against placebo and high-volume-winner anchors the surviving ordering is generic opportunity/volume geometry.

Appendix G.2. Graduation: What Ports and What Deflates

Stage by stage, Table G.1 sets each federal quantity beside its BEC analogue: what *ports*, what weakens under federal conditions, and where the structural boundary holds on both platforms. The federal CADE anchors are the same numbered adjudicated processes used on BEC, recovered through a multi-pass establishment-anchored linkage detailed in Online Supplement S5, and are *partially overlapping* with the BEC anchors by construction; the table reports portability of forensic priority, not a separate cartel population.

Table G.1: Federal Graduation: Audit-Stage Verdicts, BEC–SP versus ComprasNet

Audit stage	BEC–SP	ComprasNet	Federal verdict
Raw award-layer concentration (ROC-AUC)	0.761	0.744	Ports (pre-adjustment only)
Label-blind opportunity expectation (AUC)	0.553	0.611	Concentration ≈ opportunity
Within-stratum residual (AUC, MEDIUM cells)	0.471	0.462	Deflates on both
Matched-permutation p (within-strata shuffle)	0.127	0.906	Null not rejected
Label-frozen prospective timing (AUC)	0.713	0.595	Prospective collapse ports
Structural boundary: score vs. direct defendants (AUC)	0.491	0.651	Boundary holds (does not rank winners)
Negative-control verdict	generic geometry	generic opportunity/volume geometry	Generic geometry on both

Notes: Each row is the comparable quantity from the federal battery (Online Supplement S5) set beside its BEC analogue. The raw concentration row is reported *only* as the pre-adjustment baseline; it is opportunity arithmetic once the label-blind expectation is credited, and it is not a deployable ranking on either platform. Cobidder labels are adjudication-anchored exposure, not cartel membership; neither column ranks defendants. The federal panel has no sealed-bid modality, so no modality stratification is reported. The direct-defendant row reports point estimates; no CI band was produced for the federal scope check.

The structural boundary holds on both platforms.. The one positive cross-platform statement the graduation supports is a *negative* about the screen's reach, and it is on-thesis. On the federal panel the firm-level AUC of the loser-side score against *direct* CADE defendants is 0.651 on the full candidate universe and 0.548 within the always-loser pool (point estimates; no CI band was produced for the federal scope check), consistent with the BEC boundary of 0.491. The reason is the same on both platforms: direct defendants are predominantly winners (0.19 median win-share across the 25 federal direct-defendant firms), and a loser-side construct does not rank winners. That boundary is a feature of the construct, and it ports.

What the federal stress test does *not* return is a degraded-but-positive deployable ranking, and the appendix does not claim one. What ports is the *audit*: raw award-layer concentration looks informative until opportunity exposure is held fixed, after which the within-stratum residual deflates and the prospective ranking collapses as on BEC. The deliverable an agency carries across platforms is the decomposition that tells it how far a cheap award-layer statistic can be trusted before costly bid-layer work begins—and federally that lesson binds harder, because the bid-layer step cannot be constructed from public data at all (Section 5.4).